

GENERATIVE AI - HANDS ON - UNIT 1

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Observation Table

Task	Model	Classification	Observation	Why? (Architectural Reason)
Generation	BERT	Failure	Cannot generate; produces errors or nonsense	Encoder-only; no decoder for sequential token generation
	RoBERTa	Failure	Cannot generate; similar limitation to BERT	Encoder-only; lacks generative decoder component
	BART	Success	Generates coherent text continuations	Encoder-Decoder; decoder can autoregressively generate tokens
Fill-Mask	BERT	Success	Predicts 'create', 'generate', etc. correctly	Trained on MLM; bidirectional attention optimal for this task
	RoBERTa	Success	Similar/better performance than BERT	Trained on improved MLM; bidirectional attention

	BART	Partial Success	Predicts but less accurate than BERT/RoBERTa	Not trained on MLM; encoder-decoder not optimized for this
QA	BERT	Partial/Poor	Unreliable extraction without fine-tuning	Base model lacks QA-specific training; needs task adaptation
	RoBERTa	Partial/Poor	Similar poor performance to BERT	Base model not fine-tuned for QA; requires SQuAD training
	BART	Moderate/Poor	Generates answers instead of extracting	Generative architecture; designed for abstractive tasks, not extractive